AI-Driven Commodities Intelligence Pipeline

EXECUTIVE TECHNICAL BRIEFING | V2 ARCHITECTURE & QUANTITATIVE BACKTESTING

1. Executive Summary & Business Need

THE STRATEGIC PROBLEM

Quantitative commodities trading desks rely on capturing fundamental shifts in supply, demand, and geopolitics. However, critical macro signals are trapped in vast streams of unstructured news. Traditional human aggregation is too slow, causing desks to react to market moves rather than anticipate them.

MONETARY & RISK BENEFITS

By automating intelligence extraction with LLM agentic workflows, the pipeline ingests macro shocks and translates them into actionable quantitative indicators. This eliminates reading lag, provides auditable risk metrics, and uncovers trading alpha ahead of physical market pricing.

2. Target Audience & Stakeholders

- **Quantitative Traders & PMs:** Ingests structured Severity Scores (1-10) directly into forecasting models to capture T+3 directional momentum.
- **Chief Risk Officers (CROs):** Audits historical macro shocks against volatility spikes to optimize real-time portfolio hedging strategies.
- **Data Science & Engineering:** Serves as a modular, scalable blueprint for mitigating LLM hallucinations and deploying enterprise data pipelines.

3. Methodology & Exception Handling

The V2 architecture establishes rigorous programmatic control over probabilistic language models:

- **Data Ingestion Engine:** Automated multithreaded scrapers aggregate live global macro news via NewsAPI and continuous WTI futures price action via yfinance.
- **Agentic LLM Structuring:** Utilizes `gpt-4o-mini` with `temperature=0` and a strict Pydantic schema to extract sentiment, fundamental category, and impact severity in parallel via `ThreadPoolExecutor`.
- **Multi-Horizon Backtesting (T+3):** Evaluates directional hit rates against a 3-day forward lag, reflecting the physical time required for macroeconomic news to price into crude oil.

EXCEPTIONS & EDGE CASES HANDLED

Noise Filtering: Irrelevant macro news (e.g., tech sell-offs, Federal Reserve rate votes) is automatically classified as **"Other"** to prevent false trading triggers. **Hallucination Guardrails:** Forcing explicit reasoning fields before numerical severity assignment ensures a fully auditable decision trail.

4. Live Dashboard Exhibits (Streamlit App)

Filtered Pipeline Performance (T+3 Backtest | Severity ≥ 6)

HIGH-CONVICTION SIGNALS	CORRECT PREDICTIONS (T+3)	DIRECTIONAL HIT RATE
8	2	25.00%

Price Action vs. High-Conviction Market Fundamentals



Structured V2 AI Outputs (Sample)

Headline	Commodity	Category
Transportation crisis in oil market	Crude Oil	Supply
Wall Street post-midterm plunge	Other	Macro
Iran War inflation nightmare	Crude Oil	Macro
Oil hits 6-month high on Middle East	Crude Oil	Supply

Evaluation Ledger

Date	High-Conviction Headline
2026-10-15	Oil transportation crisis impacts costs
2026-10-19	Iran War inflation nightmare
2026-10-23	Oil prices hit 6-month high
2026-10-30	Renewed Iran war threatens Hormuz

5. Results, Analysis & Discussion

> ***Analytical Insight:*** *The 25.00% hit rate across 8 high-conviction signals reflects the extreme difficulty of short-term directional timing. However, the core engineering success lies in* ***dimensionality reduction****—filtering thousands of noisy headlines into a clean, auditable subset.*

6. Future Roadmap & Optimization

- **Enterprise Scaling:** Migrate local Python multithreading to Snowflake Snowpark Vectorized UDFs to process 50,000+ global feeds concurrently.
- **Alpha Benchmarking:** Expand evaluation metrics from directional hit rate to risk-adjusted cumulative returns (Sharpe Ratio) against WTI futures benchmarks.
- **Alternative Data:** Ingest SEC 8-K filings, earnings calls, and OPEC+ communiques into the vector extraction pipeline.